# STATIC TIMING ANALYSIS

## VSDBabySoC — SKY130 Standard-Cell Timing Corners

| | |
|---|---|
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| **Design** | VSDBabySoC |
| **Analysis** | OpenSTA Static Timing Analysis |

### 16 SKY130 PVT timing corners

# 1. Objective

To perform Static Timing Analysis (STA) of the synthesized VSDBabySoC netlist using the SKY130 HD standard-cell timing libraries across the 16 process, voltage and temperature (PVT) corners used in the assessment.

# 2. STA Methodology

The STA flow used OpenSTA with the synthesized VSDBabySoC netlist, the corresponding SKY130 timing library, the AVSDPLL and AVSDDAC libraries, and the project SDC constraints.

**Timing report command:** report_checks -path_delay min_max -fields {nets cap slew input_pins fanout}

For every corner, the minimum-delay and maximum-delay paths were reported. The screenshots in this report are the captured OpenSTA outputs for the 16 runs.

# 3. Corner Classification

FF denotes a fast-fast process corner, SS denotes a slow-slow process corner, and TT denotes a typical-typical process corner. The temperature and supply voltage are given by each library name.

# 4. STA Results Summary

| No. | Process | Temp. | VDD | Library | Min/hold slack (ns) | Max/setup slack (ns) | Status |
|---|---|---|---|---|---|---|---|
| 01 | FF | 100°C | 1.65 V | sky130_fd_sc_hd__ff_100C_1v65.lib | 0.25 | 2.35 | MET / MET |
| 02 | FF | 100°C | 1.95 V | sky130_fd_sc_hd__ff_100C_1v95.lib | 0.20 | 3.77 | MET / MET |
| 03 | FF | −40°C | 1.56 V | sky130_fd_sc_hd__ff_n40C_1v56.lib | 0.29 | 0.94 | MET / MET |
| 04 | FF | −40°C | 1.65 V | sky130_fd_sc_hd__ff_n40C_1v65.lib | 0.26 | 1.96 | MET / MET |
| 05 | FF | −40°C | 1.76 V | sky130_fd_sc_hd__ff_n40C_1v76.lib | 0.22 | 2.86 | MET / MET |
| 06 | FF | −40°C | 1.95 V | sky130_fd_sc_hd__ff_n40C_1v95.lib | 0.19 | 3.94 | MET / MET |
| 07 | SS | 100°C | 1.40 V | sky130_fd_sc_hd__ss_100C_1v40.lib | 0.91 | −13.40 | MET / VIOLATED |
| 08 | SS | 100°C | 1.60 V | sky130_fd_sc_hd__ss_100C_1v60.lib | 0.64 | −6.54 | MET / VIOLATED |
| 09 | SS | −40°C | 1.28 V | sky130_fd_sc_hd__ss_n40C_1v28.lib | 1.83 | −76.89 | MET / VIOLATED |
| 10 | SS | −40°C | 1.35 V | sky130_fd_sc_hd__ss_n40C_1v35.lib | 1.35 | −45.57 | MET / VIOLATED |
| 11 | SS | −40°C | 1.40 V | sky130_fd_sc_hd__ss_n40C_1v40.lib | 1.12 | −33.27 | MET / VIOLATED |
| 12 | SS | −40°C | 1.44 V | sky130_fd_sc_hd__ss_n40C_1v44.lib | 0.99 | −27.31 | MET / VIOLATED |
| 13 | SS | −40°C | 1.60 V | sky130_fd_sc_hd__ss_n40C_1v60.lib | 0.66 | −11.18 | MET / VIOLATED |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 14 | SS | −40°C | 1.76 V | sky130_fd_sc_hd__ss_n40C_1v76.lib | 0.50 | −4.76 | MET / VIOLATED |
| 15 | TT | 25°C | 1.80 V | sky130_fd_sc_hd__tt_025C_1v80.lib | 0.31 | 0.96 | MET / MET |
| 16 | TT | 100°C | 1.80 V | sky130_fd_sc_hd__tt_100C_1v80.lib | 0.31 | 1.02 | MET / MET |

Note: A negative maximum-path slack is a setup-timing violation at that PVT corner. It is an STA result and is reported as observed; it is not treated as a documentation error.

# 5.01 sky130_fd_sc_hd__ff_100C_1v65.lib

**PVT condition:** FF process, 100°C, 1.65 V supply

Fast-fast process corner. The two transistor/cell sides are characterized as fast.

**Observed STA result:** minimum-path slack = 0.25 ns (MET); maximum-path slack = 2.35 ns (MET).

*OpenSTA report screenshot*

```
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _9423_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.03    0.22    0.22 ^ _9423_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           core.CPU_Xreg_value_a5[17][0] (net)
                  0.03    0.00    0.22 ^ _10265_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.22   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ _10265_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.03   -0.03   library hold time
                                 -0.03   data required time
---------------------------------------------------------------------------
                                 -0.03   data required time
                                 -0.22   data arrival time
---------------------------------------------------------------------------
                                  0.25   slack (MET)


Startpoint: _10392_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10042_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _10392_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  4.91    3.56    3.56 ^ _10392_/Q (sky130_fd_sc_hd__dfxtp_1)
   253    0.63                           core.CPU_reset_a4 (net)
                  4.91    0.00    3.56 ^ _8062_/A (sky130_fd_sc_hd__clkinv_1)
                  2.21    4.55    8.12 v _8062_/Y (sky130_fd_sc_hd__clkinv_1)
   260    0.45                           _3834_ (net)
                  2.21    0.00    8.12 v _8767_/B1 (sky130_fd_sc_hd__o211ai_1)
                  0.31    0.44    8.56 ^ _8767_/Y (sky130_fd_sc_hd__o211ai_1)
     1    0.00                           _683_ (net)
                  0.31    0.00    8.56 ^ _10042_/D (sky130_fd_sc_hd__dfxtp_1)
                                  8.56   data arrival time

                  0.00   11.00   11.00   clock clk (rise edge)
                          0.00   11.00   clock network delay (ideal)
                          0.00   11.00   clock reconvergence pessimism
                                 11.00 ^ _10042_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.10   10.90   library setup time
                                 10.90   data required time
---------------------------------------------------------------------------
                                 10.90   data required time
                                 -8.56   data arrival time
---------------------------------------------------------------------------
                                  2.35   slack (MET)
```

# 5.02 sky130_fd_sc_hd__ff_100C_1v95.lib

**PVT condition:** FF process, 100°C, 1.95 V supply

Fast-fast process corner. The two transistor/cell sides are characterized as fast.

**Observed STA result:** minimum-path slack = 0.20 ns (MET); maximum-path slack = 3.77 ns (MET).

*OpenSTA report screenshot*

```
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap     Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _9423_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.03    0.18    0.18 ^ _9423_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           core.CPU_Xreg_value_a5[17][0] (net)
                  0.03    0.00    0.18 ^ _10265_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.18   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ _10265_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.02   -0.02   library hold time
                                 -0.02   data required time
---------------------------------------------------------------------------
                                 -0.02   data required time
                                 -0.18   data arrival time
---------------------------------------------------------------------------
                                  0.20   slack (MET)


Startpoint: _10392_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10042_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap     Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _10392_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  4.15    3.00    3.00 ^ _10392_/Q (sky130_fd_sc_hd__dfxtp_1)
   253    0.65                           core.CPU_reset_a4 (net)
                  4.15    0.00    3.00 ^ _8062_/A (sky130_fd_sc_hd__clkinv_1)
                  2.42    3.73    6.73 v _8062_/Y (sky130_fd_sc_hd__clkinv_1)
   260    0.47                           _3834_ (net)
                  2.42    0.00    6.73 v _8767_/B1 (sky130_fd_sc_hd__o211ai_1)
                  0.31    0.44    7.17 ^ _8767_/Y (sky130_fd_sc_hd__o211ai_1)
     1    0.00                           _683_ (net)
                  0.31    0.00    7.17 ^ _10042_/D (sky130_fd_sc_hd__dfxtp_1)
                                  7.17   data arrival time

                  0.00   11.00   11.00   clock clk (rise edge)
                          0.00   11.00   clock network delay (ideal)
                          0.00   11.00   clock reconvergence pessimism
                                 11.00 ^ _10042_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.06   10.94   library setup time
                                 10.94   data required time
---------------------------------------------------------------------------
                                 10.94   data required time
                                 -7.17   data arrival time
---------------------------------------------------------------------------
                                  3.77   slack (MET)
```

# 5.03 sky130_fd_sc_hd__ff_n40C_1v56.lib

**PVT condition:** FF process, −40°C, 1.56 V supply

Fast-fast process corner. The two transistor/cell sides are characterized as fast.

**Observed STA result:** minimum-path slack = 0.29 ns (MET); maximum-path slack = 0.94 ns (MET).

*OpenSTA report screenshot*

```
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _9423_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.03    0.26    0.26 ^ _9423_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           core.CPU_Xreg_value_a5[17][0] (net)
                  0.03    0.00    0.26 ^ _10265_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.26   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ _10265_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.03   -0.03   library hold time
                                 -0.03   data required time
-----------------------------------------------------------------------------
                                 -0.03   data required time
                                 -0.26   data arrival time
-----------------------------------------------------------------------------
                                  0.29   slack (MET)


Startpoint: _10392_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10042_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _10392_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  6.22    4.58    4.58 ^ _10392_/Q (sky130_fd_sc_hd__dfxtp_1)
   253    0.59                           core.CPU_reset_a4 (net)
                  6.22    0.00    4.58 ^ _8062_/A (sky130_fd_sc_hd__clkinv_1)
                  1.57    4.72    9.30 v _8062_/Y (sky130_fd_sc_hd__clkinv_1)
   260    0.42                           _3834_ (net)
                  1.57    0.00    9.30 v _8767_/B1 (sky130_fd_sc_hd__o211ai_1)
                  0.29    0.64    9.94 ^ _8767_/Y (sky130_fd_sc_hd__o211ai_1)
     1    0.00                           _683_ (net)
                  0.29    0.00    9.94 ^ _10042_/D (sky130_fd_sc_hd__dfxtp_1)
                                  9.94   data arrival time

                  0.00   11.00   11.00   clock clk (rise edge)
                          0.00   11.00   clock network delay (ideal)
                          0.00   11.00   clock reconvergence pessimism
                                 11.00 ^ _10042_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.12   10.88   library setup time
                                 10.88   data required time
-----------------------------------------------------------------------------
                                 10.88   data required time
                                 -9.94   data arrival time
-----------------------------------------------------------------------------
                                  0.94   slack (MET)
```

# 5.04 sky130_fd_sc_hd__ff_n40C_1v65.lib

**PVT condition:** FF process, −40°C, 1.65 V supply

Fast-fast process corner. The two transistor/cell sides are characterized as fast.

**Observed STA result:** minimum-path slack = 0.26 ns (MET); maximum-path slack = 1.96 ns (MET).

*OpenSTA report screenshot*

```
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _9423_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.03    0.23    0.23 ^ _9423_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           core.CPU_Xreg_value_a5[17][0] (net)
                  0.03    0.00    0.23 ^ _10265_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.23   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ _10265_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.02   -0.02   library hold time
                                 -0.02   data required time
---------------------------------------------------------------------------
                                 -0.02   data required time
                                 -0.23   data arrival time
---------------------------------------------------------------------------
                                  0.26   slack (MET)


Startpoint: _10392_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10042_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _10392_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  5.64    4.10    4.10 ^ _10392_/Q (sky130_fd_sc_hd__dfxtp_1)
   253    0.60                           core.CPU_reset_a4 (net)
                  5.64    0.00    4.10 ^ _8062_/A (sky130_fd_sc_hd__clkinv_1)
                  1.60    4.27    8.37 v _8062_/Y (sky130_fd_sc_hd__clkinv_1)
   260    0.42                           _3834_ (net)
                  1.60    0.00    8.37 v _8767_/B1 (sky130_fd_sc_hd__o211ai_1)
                  0.28    0.57    8.94 ^ _8767_/Y (sky130_fd_sc_hd__o211ai_1)
     1    0.00                           _683_ (net)
                  0.28    0.00    8.94 ^ _10042_/D (sky130_fd_sc_hd__dfxtp_1)
                                  8.94   data arrival time

                  0.00   11.00   11.00   clock clk (rise edge)
                          0.00   11.00   clock network delay (ideal)
                          0.00   11.00   clock reconvergence pessimism
                                 11.00 ^ _10042_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.10   10.90   library setup time
                                 10.90   data required time
---------------------------------------------------------------------------
                                 10.90   data required time
                                 -8.94   data arrival time
---------------------------------------------------------------------------
                                  1.96   slack (MET)
```

# 5.05 sky130_fd_sc_hd__ff_n40C_1v76.lib

**PVT condition:** FF process, −40°C, 1.76 V supply

Fast-fast process corner. The two transistor/cell sides are characterized as fast.

**Observed STA result:** minimum-path slack = 0.22 ns (MET); maximum-path slack = 2.86 ns (MET).

*OpenSTA report screenshot*

```
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _9423_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.03    0.20    0.20 ^ _9423_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           core.CPU_Xreg_value_a5[17][0] (net)
                  0.03    0.00    0.20 ^ _10265_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.20   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ _10265_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.02   -0.02   library hold time
                                 -0.02   data required time
-----------------------------------------------------------------------------
                                 -0.02   data required time
                                 -0.20   data arrival time
-----------------------------------------------------------------------------
                                  0.22   slack (MET)


Startpoint: _10392_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10042_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _10392_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  5.03    3.71    3.71 ^ _10392_/Q (sky130_fd_sc_hd__dfxtp_1)
   253    0.62                           core.CPU_reset_a4 (net)
                  5.03    0.00    3.71 ^ _8062_/A (sky130_fd_sc_hd__clkinv_1)
                  1.79    3.82    7.53 v _8062_/Y (sky130_fd_sc_hd__clkinv_1)
   260    0.43                           _3834_ (net)
                  1.79    0.00    7.53 v _8767_/B1 (sky130_fd_sc_hd__o211ai_1)
                  0.27    0.54    8.07 ^ _8767_/Y (sky130_fd_sc_hd__o211ai_1)
     1    0.00                           _683_ (net)
                  0.27    0.00    8.07 ^ _10042_/D (sky130_fd_sc_hd__dfxtp_1)
                                  8.07   data arrival time

                  0.00   11.00   11.00   clock clk (rise edge)
                          0.00   11.00   clock network delay (ideal)
                          0.00   11.00   clock reconvergence pessimism
                                 11.00 ^ _10042_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.08   10.92   library setup time
                                 10.92   data required time
-----------------------------------------------------------------------------
                                 10.92   data required time
                                 -8.07   data arrival time
-----------------------------------------------------------------------------
                                  2.86   slack (MET)
```

# 5.06 sky130_fd_sc_hd__ff_n40C_1v95.lib

**PVT condition:** FF process, −40°C, 1.95 V supply

Fast-fast process corner. The two transistor/cell sides are characterized as fast.

**Observed STA result:** minimum-path slack = 0.19 ns (MET); maximum-path slack = 3.94 ns (MET).

*OpenSTA report screenshot*

```
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _9423_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.02    0.17    0.17 ^ _9423_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           core.CPU_Xreg_value_a5[17][0] (net)
                  0.02    0.00    0.17 ^ _10265_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.17   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ _10265_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.02   -0.02   library hold time
                                 -0.02   data required time
---------------------------------------------------------------------------
                                 -0.02   data required time
                                 -0.17   data arrival time
---------------------------------------------------------------------------
                                  0.19   slack (MET)


Startpoint: _10392_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10042_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _10392_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  4.45    3.20    3.20 ^ _10392_/Q (sky130_fd_sc_hd__dfxtp_1)
   253    0.63                           core.CPU_reset_a4 (net)
                  4.45    0.00    3.20 ^ _8062_/A (sky130_fd_sc_hd__clkinv_1)
                  1.89    3.33    6.53 v _8062_/Y (sky130_fd_sc_hd__clkinv_1)
   260    0.45                           _3834_ (net)
                  1.89    0.00    6.53 v _8767_/B1 (sky130_fd_sc_hd__o211ai_1)
                  0.24    0.49    7.01 ^ _8767_/Y (sky130_fd_sc_hd__o211ai_1)
     1    0.00                           _683_ (net)
                  0.24    0.00    7.01 ^ _10042_/D (sky130_fd_sc_hd__dfxtp_1)
                                  7.01   data arrival time

                  0.00   11.00   11.00   clock clk (rise edge)
                          0.00   11.00   clock network delay (ideal)
                          0.00   11.00   clock reconvergence pessimism
                                 11.00 ^ _10042_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.05   10.95   library setup time
                                 10.95   data required time
---------------------------------------------------------------------------
                                 10.95   data required time
                                 -7.01   data arrival time
---------------------------------------------------------------------------
                                  3.94   slack (MET)
```

*Figure 06: OpenSTA min/max path report for sky130_fd_sc_hd__ff_n40C_1v95.lib*

# 5.07 sky130_fd_sc_hd__ss_100C_1v40.lib

**PVT condition:** SS process, 100°C, 1.40 V supply

Slow-slow process corner. The two transistor/cell sides are characterized as slow.

**Observed STA result:** minimum-path slack = 0.91 ns (MET); maximum-path slack = −13.40 ns (VIOLATED).

*OpenSTA report screenshot*

```
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _9423_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.05    0.75    0.75 v _9423_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           core.CPU_Xreg_value_a5[17][0] (net)
                  0.05    0.00    0.75 v _10265_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.75   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ _10265_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.15   -0.15   library hold time
                                 -0.15   data required time
-----------------------------------------------------------------------------
                                 -0.15   data required time
                                 -0.75   data arrival time
-----------------------------------------------------------------------------
                                  0.91   slack (MET)


Startpoint: _10392_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10233_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _10392_/CLK (sky130_fd_sc_hd__dfxtp_1)
                 11.68    9.24    9.24 ^ _10392_/Q (sky130_fd_sc_hd__dfxtp_1)
   253    0.60                           core.CPU_reset_a4 (net)
                 11.68    0.00    9.24 ^ _8062_/A (sky130_fd_sc_hd__clkinv_1)
                  8.21   12.23   21.47 v _8062_/Y (sky130_fd_sc_hd__clkinv_1)
   260    0.41                           _3834_ (net)
                  8.21    0.00   21.47 v _9277_/B1 (sky130_fd_sc_hd__o21ai_0)
                  1.44    1.87   23.34 ^ _9277_/Y (sky130_fd_sc_hd__o21ai_0)
     1    0.00                           _4573_ (net)
                  1.44    0.00   23.34 ^ _9278_/B1 (sky130_fd_sc_hd__a21oi_1)
                  0.36    0.45   23.79 v _9278_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _874_ (net)
                  0.36    0.00   23.79 v _10233_/D (sky130_fd_sc_hd__dfxtp_1)
                                 23.79   data arrival time

                  0.00   11.00   11.00   clock clk (rise edge)
                          0.00   11.00   clock network delay (ideal)
                          0.00   11.00   clock reconvergence pessimism
                                 11.00 ^ _10233_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.61   10.39   library setup time
                                 10.39   data required time
-----------------------------------------------------------------------------
                                 10.39   data required time
                                -23.79   data arrival time
-----------------------------------------------------------------------------
                                -13.40   slack (VIOLATED)
```

# 5.08 sky130_fd_sc_hd__ss_100C_1v60.lib

**PVT condition:** SS process, 100°C, 1.60 V supply

Slow-slow process corner. The two transistor/cell sides are characterized as slow.

**Observed STA result:** minimum-path slack = 0.64 ns (MET); maximum-path slack = −6.54 ns (VIOLATED).

*OpenSTA report screenshot*

```
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap     Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _9423_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.04    0.53    0.53 v _9423_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           core.CPU_Xreg_value_a5[17][0] (net)
                  0.04    0.00    0.53 v _10265_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.53   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ _10265_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.11   -0.11   library hold time
                                 -0.11   data required time
---------------------------------------------------------------------------
                                 -0.11   data required time
                                 -0.53   data arrival time
---------------------------------------------------------------------------
                                  0.64   slack (MET)


Startpoint: _10392_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10233_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap     Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _10392_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  9.00    6.95    6.95 ^ _10392_/Q (sky130_fd_sc_hd__dfxtp_1)
   253    0.62                           core.CPU_reset_a4 (net)
                  9.00    0.00    6.95 ^ _8062_/A (sky130_fd_sc_hd__clkinv_1)
                  5.81    8.89   15.84 v _8062_/Y (sky130_fd_sc_hd__clkinv_1)
   260    0.43                           _3834_ (net)
                  5.81    0.00   15.84 v _9277_/B1 (sky130_fd_sc_hd__o21ai_0)
                  1.00    1.08   16.92 ^ _9277_/Y (sky130_fd_sc_hd__o21ai_0)
     1    0.00                           _4573_ (net)
                  1.00    0.00   16.92 ^ _9278_/B1 (sky130_fd_sc_hd__a21oi_1)
                  0.29    0.24   17.16 v _9278_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _874_ (net)
                  0.29    0.00   17.16 v _10233_/D (sky130_fd_sc_hd__dfxtp_1)
                                 17.16   data arrival time

                  0.00   11.00   11.00   clock clk (rise edge)
                          0.00   11.00   clock network delay (ideal)
                          0.00   11.00   clock reconvergence pessimism
                                 11.00 ^ _10233_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.38   10.62   library setup time
                                 10.62   data required time
---------------------------------------------------------------------------
                                 10.62   data required time
                                -17.16   data arrival time
---------------------------------------------------------------------------
                                 -6.54   slack (VIOLATED)
```

# 5.09 sky130_fd_sc_hd__ss_n40C_1v28.lib

**PVT condition:** SS process, −40°C, 1.28 V supply

Slow-slow process corner. The two transistor/cell sides are characterized as slow.

**Observed STA result:** minimum-path slack = 1.83 ns (MET); maximum-path slack = −76.89 ns (VIOLATED).

*OpenSTA report screenshot*

```
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _9423_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.07    1.45    1.45 v _9423_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           core.CPU_Xreg_value_a5[17][0] (net)
                  0.07    0.00    1.45 v _10265_/D (sky130_fd_sc_hd__dfxtp_1)
                                  1.45   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ _10265_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.38   -0.38   library hold time
                                 -0.38   data required time
---------------------------------------------------------------------------
                                 -0.38   data required time
                                 -1.45   data arrival time
---------------------------------------------------------------------------
                                  1.83   slack (MET)


Startpoint: _10277_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10321_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _10277_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.10    1.49    1.49 v _10277_/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.00                           core.CPU_valid_load_a5 (net)
                  0.10    0.00    1.49 v _4607_/A (sky130_fd_sc_hd__nor4_1)
                 25.67   21.42   22.92 ^ _4607_/Y (sky130_fd_sc_hd__nor4_1)
    39    0.08                           core.CPU_valid_a3 (net)
                 25.67    0.00   22.92 ^ _4821_/A3 (sky130_fd_sc_hd__a31oi_1)
                  1.91   12.13   35.05 v _4821_/Y (sky130_fd_sc_hd__a31oi_1)
     3    0.01                           _1096_ (net)
                  1.91    0.00   35.05 v _6259_/D (sky130_fd_sc_hd__nor4_1)
                 38.72   32.19   67.24 ^ _6259_/Y (sky130_fd_sc_hd__nor4_1)
    63    0.11                           _2498_ (net)
                 38.72    0.00   67.24 ^ _6286_/A1 (sky130_fd_sc_hd__o21ai_0)
                  2.49   17.40   84.64 v _6286_/Y (sky130_fd_sc_hd__o21ai_0)
     1    0.00                           core.CPU_src2_value_a2[1] (net)
                  2.49    0.00   84.64 v _10321_/D (sky130_fd_sc_hd__dfxtp_1)
                                 84.64   data arrival time

                  0.00   11.00   11.00   clock clk (rise edge)
                          0.00   11.00   clock network delay (ideal)
                          0.00   11.00   clock reconvergence pessimism
                                 11.00 ^ _10321_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -3.25    7.75   library setup time
                                  7.75   data required time
---------------------------------------------------------------------------
                                  7.75   data required time
                                -84.64   data arrival time
---------------------------------------------------------------------------
                                -76.89   slack (VIOLATED)
```

## 5.10 sky130_fd_sc_hd__ss_n40C_1v35.lib

**PVT condition:** SS process, −40°C, 1.35 V supply

Slow-slow process corner. The two transistor/cell sides are characterized as slow.

**Observed STA result:** minimum-path slack = 1.35 ns (MET); maximum-path slack = −45.57 ns (VIOLATED).

*OpenSTA report screenshot*

```
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _9423_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.06    1.08    1.08 v _9423_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           core.CPU_Xreg_value_a5[17][0] (net)
                  0.06    0.00    1.08 v _10265_/D (sky130_fd_sc_hd__dfxtp_1)
                                  1.08   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ _10265_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.27   -0.27   library hold time
                                 -0.27   data required time
-----------------------------------------------------------------------------
                                 -0.27   data required time
                                 -1.08   data arrival time
-----------------------------------------------------------------------------
                                  1.35   slack (MET)


Startpoint: _10277_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10321_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _10277_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.08    1.11    1.11 v _10277_/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.00                           core.CPU_valid_load_a5 (net)
                  0.08    0.00    1.11 v _4607_/A (sky130_fd_sc_hd__nor4_1)
                 17.79   14.84   15.95 ^ _4607_/Y (sky130_fd_sc_hd__nor4_1)
    39    0.08                           core.CPU_valid_a3 (net)
                 17.79    0.00   15.95 ^ _4821_/A3 (sky130_fd_sc_hd__a31oi_1)
                  1.44    6.86   22.81 v _4821_/Y (sky130_fd_sc_hd__a31oi_1)
     3    0.01                           _1096_ (net)
                  1.44    0.00   22.81 v _6259_/D (sky130_fd_sc_hd__nor4_1)
                 26.54   22.25   45.06 ^ _6259_/Y (sky130_fd_sc_hd__nor4_1)
    63    0.12                           _2498_ (net)
                 26.54    0.00   45.06 ^ _6286_/A1 (sky130_fd_sc_hd__o21ai_0)
                  1.72    9.39   54.45 v _6286_/Y (sky130_fd_sc_hd__o21ai_0)
     1    0.00                           core.CPU_src2_value_a2[1] (net)
                  1.72    0.00   54.45 v _10321_/D (sky130_fd_sc_hd__dfxtp_1)
                                 54.45   data arrival time

                  0.00   11.00   11.00   clock clk (rise edge)
                          0.00   11.00   clock network delay (ideal)
                          0.00   11.00   clock reconvergence pessimism
                                 11.00 ^ _10321_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -2.13    8.87   library setup time
                                  8.87   data required time
-----------------------------------------------------------------------------
                                  8.87   data required time
                                -54.45   data arrival time
-----------------------------------------------------------------------------
                                -45.57   slack (VIOLATED)
```

## 5.11 sky130_fd_sc_hd__ss_n40C_1v40.lib

**PVT condition:** SS process, −40°C, 1.40 V supply

Slow-slow process corner. The two transistor/cell sides are characterized as slow.

**Observed STA result:** minimum-path slack = 1.12 ns (MET); maximum-path slack = −33.27 ns (VIOLATED).

*OpenSTA report screenshot*

```
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _9423_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.05    0.91    0.91 v _9423_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           core.CPU_Xreg_value_a5[17][0] (net)
                  0.05    0.00    0.91 v _10265_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.91   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ _10265_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.22   -0.22   library hold time
                                 -0.22   data required time
-----------------------------------------------------------------------------
                                 -0.22   data required time
                                 -0.91   data arrival time
-----------------------------------------------------------------------------
                                  1.12   slack (MET)


Startpoint: _10277_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10321_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _10277_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.07    0.94    0.94 v _10277_/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.00                           core.CPU_valid_load_a5 (net)
                  0.07    0.00    0.94 v _4607_/A (sky130_fd_sc_hd__nor4_1)
                 14.35   11.95   12.88 ^ _4607_/Y (sky130_fd_sc_hd__nor4_1)
    39    0.08                           core.CPU_valid_a3 (net)
                 14.35    0.00   12.88 ^ _4821_/A3 (sky130_fd_sc_hd__a31oi_1)
                  1.18    5.07   17.95 v _4821_/Y (sky130_fd_sc_hd__a31oi_1)
     3    0.01                           _1096_ (net)
                  1.18    0.00   17.95 v _6259_/D (sky130_fd_sc_hd__nor4_1)
                 21.38   17.86   35.81 ^ _6259_/Y (sky130_fd_sc_hd__nor4_1)
    63    0.12                           _2498_ (net)
                 21.38    0.00   35.81 ^ _6286_/A1 (sky130_fd_sc_hd__o21ai_0)
                  1.40    6.81   42.62 v _6286_/Y (sky130_fd_sc_hd__o21ai_0)
     1    0.00                           core.CPU_src2_value_a2[1] (net)
                  1.40    0.00   42.62 v _10321_/D (sky130_fd_sc_hd__dfxtp_1)
                                 42.62   data arrival time

                  0.00   11.00   11.00   clock clk (rise edge)
                          0.00   11.00   clock network delay (ideal)
                          0.00   11.00   clock reconvergence pessimism
                                 11.00 ^ _10321_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -1.65    9.35   library setup time
                                  9.35   data required time
-----------------------------------------------------------------------------
                                  9.35   data required time
                                -42.62   data arrival time
-----------------------------------------------------------------------------
                                -33.27   slack (VIOLATED)
```

## 5.12 sky130_fd_sc_hd__ss_n40C_1v44.lib

**PVT condition:** SS process, −40°C, 1.44 V supply

Slow-slow process corner. The two transistor/cell sides are characterized as slow.

**Observed STA result:** minimum-path slack = 0.99 ns (MET); maximum-path slack = −27.31 ns (VIOLATED).

*OpenSTA report screenshot*

```
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _9423_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.05    0.80    0.80 v _9423_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           core.CPU_Xreg_value_a5[17][0] (net)
                  0.05    0.00    0.80 v _10265_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.80   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ _10265_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.19   -0.19   library hold time
                                 -0.19   data required time
---------------------------------------------------------------------------
                                 -0.19   data required time
                                 -0.80   data arrival time
---------------------------------------------------------------------------
                                  0.99   slack (MET)


Startpoint: _10277_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10321_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _10277_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.07    0.83    0.83 v _10277_/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.00                           core.CPU_valid_load_a5 (net)
                  0.07    0.00    0.83 v _4607_/A (sky130_fd_sc_hd__nor4_1)
                 12.26   10.23   11.05 ^ _4607_/Y (sky130_fd_sc_hd__nor4_1)
    39    0.08                           core.CPU_valid_a3 (net)
                 12.26    0.00   11.05 ^ _4821_/A3 (sky130_fd_sc_hd__a31oi_1)
                  1.05    4.43   15.48 v _4821_/Y (sky130_fd_sc_hd__a31oi_1)
     3    0.01                           _1096_ (net)
                  1.05    0.00   15.48 v _6259_/D (sky130_fd_sc_hd__nor4_1)
                 18.30   15.43   30.91 ^ _6259_/Y (sky130_fd_sc_hd__nor4_1)
    63    0.12                           _2498_ (net)
                 18.30    0.00   30.91 ^ _6286_/A1 (sky130_fd_sc_hd__o21ai_0)
                  1.25    5.99   36.89 v _6286_/Y (sky130_fd_sc_hd__o21ai_0)
     1    0.00                           core.CPU_src2_value_a2[1] (net)
                  1.25    0.00   36.89 v _10321_/D (sky130_fd_sc_hd__dfxtp_1)
                                 36.89   data arrival time

                  0.00   11.00   11.00   clock clk (rise edge)
                          0.00   11.00   clock network delay (ideal)
                          0.00   11.00   clock reconvergence pessimism
                                 11.00 ^ _10321_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -1.42    9.58   library setup time
                                  9.58   data required time
---------------------------------------------------------------------------
                                  9.58   data required time
                                -36.89   data arrival time
---------------------------------------------------------------------------
                                -27.31   slack (VIOLATED)
```

## 5.13 sky130_fd_sc_hd__ss_n40C_1v60.lib

**PVT condition:** SS process, −40°C, 1.60 V supply

Slow-slow process corner. The two transistor/cell sides are characterized as slow.

**Observed STA result:** minimum-path slack = 0.66 ns (MET); maximum-path slack = −11.18 ns (VIOLATED).

*OpenSTA report screenshot*

```
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _9423_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.03    0.54    0.54 v _9423_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           core.CPU_Xreg_value_a5[17][0] (net)
                  0.03    0.00    0.54 v _10265_/D (sky130_fd_sc_hd__dfxtp_1)
                          0.54    0.54   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ _10265_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.12   -0.12   library hold time
                                 -0.12   data required time
---------------------------------------------------------------------------
                                 -0.12   data required time
                                 -0.54   data arrival time
---------------------------------------------------------------------------
                                  0.66   slack (MET)


Startpoint: _10277_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10321_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _10277_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.05    0.56    0.56 v _10277_/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.00                           core.CPU_valid_load_a5 (net)
                  0.05    0.00    0.56 v _4607_/A (sky130_fd_sc_hd__nor4_1)
                  7.82    6.42    6.98 ^ _4607_/Y (sky130_fd_sc_hd__nor4_1)
    39    0.09                           core.CPU_valid_a3 (net)
                  7.82    0.00    6.98 ^ _4821_/A3 (sky130_fd_sc_hd__a31oi_1)
                  0.73    2.12    9.11 v _4821_/Y (sky130_fd_sc_hd__a31oi_1)
     3    0.01                           _1096_ (net)
                  0.73    0.00    9.11 v _6259_/D (sky130_fd_sc_hd__nor4_1)
                 11.67    9.57   18.68 ^ _6259_/Y (sky130_fd_sc_hd__nor4_1)
    63    0.13                           _2498_ (net)
                 11.67    0.00   18.68 ^ _6286_/A1 (sky130_fd_sc_hd__o21ai_0)
                  0.83    2.69   21.37 v _6286_/Y (sky130_fd_sc_hd__o21ai_0)
     1    0.00                           core.CPU_src2_value_a2[1] (net)
                  0.83    0.00   21.37 v _10321_/D (sky130_fd_sc_hd__dfxtp_1)
                          21.37   21.37   data arrival time

                  0.00   11.00   11.00   clock clk (rise edge)
                          0.00   11.00   clock network delay (ideal)
                          0.00   11.00   clock reconvergence pessimism
                                 11.00 ^ _10321_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.81   10.19   library setup time
                                 10.19   data required time
---------------------------------------------------------------------------
                                 10.19   data required time
                                -21.37   data arrival time
---------------------------------------------------------------------------
                                -11.18   slack (VIOLATED)
```

## 5.14 sky130_fd_sc_hd__ss_n40C_1v76.lib

**PVT condition:** SS process, −40°C, 1.76 V supply

Slow-slow process corner. The two transistor/cell sides are characterized as slow.

**Observed STA result:** minimum-path slack = 0.50 ns (MET); maximum-path slack = −4.76 ns (VIOLATED).

*OpenSTA report screenshot*

```
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _9423_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.03    0.41    0.41 v _9423_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           core.CPU_Xreg_value_a5[17][0] (net)
                  0.03    0.00    0.41 v _10265_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.41   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                          0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ _10265_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.09   -0.09   library hold time
                                 -0.09   data required time
-----------------------------------------------------------------------------
                                 -0.09   data required time
                                 -0.41   data arrival time
-----------------------------------------------------------------------------
                                  0.50   slack (MET)


Startpoint: _10277_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10321_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
-----------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                          0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _10277_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.04    0.43    0.43 v _10277_/Q (sky130_fd_sc_hd__dfxtp_1)
     2    0.00                           core.CPU_valid_load_a5 (net)
                  0.04    0.00    0.43 v _4607_/A (sky130_fd_sc_hd__nor4_1)
                  5.65    4.59    5.02 ^ _4607_/Y (sky130_fd_sc_hd__nor4_1)
    39    0.09                           core.CPU_valid_a3 (net)
                  5.65    0.00    5.02 ^ _4821_/A3 (sky130_fd_sc_hd__a31oi_1)
                  0.60    1.47    6.48 v _4821_/Y (sky130_fd_sc_hd__a31oi_1)
     3    0.01                           _1096_ (net)
                  0.60    0.00    6.48 v _6259_/D (sky130_fd_sc_hd__nor4_1)
                  8.45    6.84   13.32 ^ _6259_/Y (sky130_fd_sc_hd__nor4_1)
    63    0.13                           _2498_ (net)
                  8.45    0.00   13.32 ^ _6286_/A1 (sky130_fd_sc_hd__o21ai_0)
                  0.73    1.83   15.16 v _6286_/Y (sky130_fd_sc_hd__o21ai_0)
     1    0.00                           core.CPU_src2_value_a2[1] (net)
                  0.73    0.00   15.16 v _10321_/D (sky130_fd_sc_hd__dfxtp_1)
                                 15.16   data arrival time

                  0.00   11.00   11.00   clock clk (rise edge)
                          0.00   11.00   clock network delay (ideal)
                          0.00   11.00   clock reconvergence pessimism
                                 11.00 ^ _10321_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.61   10.39   library setup time
                                 10.39   data required time
-----------------------------------------------------------------------------
                                 10.39   data required time
                                -15.16   data arrival time
-----------------------------------------------------------------------------
                                 -4.76   slack (VIOLATED)
```

## 5.15 sky130_fd_sc_hd__tt_025C_1v80.lib

**PVT condition:** TT process, 25°C, 1.80 V supply

Typical-typical process corner. The cells are characterized at the typical process condition.

**Observed STA result:** minimum-path slack = 0.31 ns (MET); maximum-path slack = 0.96 ns (MET).

*OpenSTA report screenshot*

```
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                  0.00    0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _9423_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.03    0.27    0.27 ^ _9423_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           core.CPU_Xreg_value_a5[17][0] (net)
                  0.03    0.00    0.27 ^ _10265_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.27   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                  0.00    0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ _10265_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.03   -0.03   library hold time
                                 -0.03   data required time
---------------------------------------------------------------------------
                                 -0.03   data required time
                                 -0.27   data arrival time
---------------------------------------------------------------------------
                                  0.31   slack (MET)


Startpoint: _10392_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10042_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap    Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                  0.00    0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _10392_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  5.77    4.31    4.31 ^ _10392_/Q (sky130_fd_sc_hd__dfxtp_1)
   253    0.63                           core.CPU_reset_a4 (net)
                  5.77    0.00    4.31 ^ _8062_/A (sky130_fd_sc_hd__clkinv_1)
                  1.95    5.05    9.36 v _8062_/Y (sky130_fd_sc_hd__clkinv_1)
   260    0.43                           _3834_ (net)
                  1.95    0.00    9.36 v _8767_/B1 (sky130_fd_sc_hd__o211ai_1)
                  0.37    0.55    9.91 ^ _8767_/Y (sky130_fd_sc_hd__o211ai_1)
     1    0.00                           _683_ (net)
                  0.37    0.00    9.91 ^ _10042_/D (sky130_fd_sc_hd__dfxtp_1)
                                  9.91   data arrival time

                  0.00   11.00   11.00   clock clk (rise edge)
                         0.00    11.00   clock network delay (ideal)
                         0.00    11.00   clock reconvergence pessimism
                                 11.00 ^ _10042_/CLK (sky130_fd_sc_hd__dfxtp_1)
                        -0.14    10.86   library setup time
                                 10.86   data required time
---------------------------------------------------------------------------
                                 10.86   data required time
                                 -9.91   data arrival time
---------------------------------------------------------------------------
                                  0.96   slack (MET)
```

## 5.16 sky130_fd_sc_hd__tt_100C_1v80.lib

**PVT condition:** TT process, 100°C, 1.80 V supply

Typical-typical process corner. The cells are characterized at the typical process condition.

**Observed STA result:** minimum-path slack = 0.31 ns (MET); maximum-path slack = 1.02 ns (MET).

*OpenSTA report screenshot*

```
Warning: /home/vsduser/Desktop/vsdbabysoc/src/ltd/avsddac.ltd line 1, library avsddac alrea
Startpoint: _9423_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10265_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Fanout     Cap     Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                  0.00    0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _9423_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  0.03    0.28    0.28 ^ _9423_/Q (sky130_fd_sc_hd__dfxtp_1)
     1    0.00                           core.CPU_Xreg_value_a5[17][0] (net)
                  0.03    0.00    0.28 ^ _10265_/D (sky130_fd_sc_hd__dfxtp_1)
                                  0.28   data arrival time

                  0.00    0.00    0.00   clock clk (rise edge)
                  0.00    0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00   clock reconvergence pessimism
                                  0.00 ^ _10265_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.04   -0.04   library hold time
                                 -0.04   data required time
---------------------------------------------------------------------------
                                 -0.04   data required time
                                 -0.28   data arrival time
---------------------------------------------------------------------------
                                  0.31   slack (MET)


Startpoint: _10392_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _10233_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout     Cap     Slew   Delay    Time   Description
---------------------------------------------------------------------------
                  0.00    0.00    0.00   clock clk (rise edge)
                  0.00    0.00    0.00   clock network delay (ideal)
                  0.00    0.00    0.00 ^ _10392_/CLK (sky130_fd_sc_hd__dfxtp_1)
                  5.56    4.07    4.07 ^ _10392_/Q (sky130_fd_sc_hd__dfxtp_1)
   253    0.64                           core.CPU_reset_a4 (net)
                  5.56    0.00    4.07 ^ _8062_/A (sky130_fd_sc_hd__clkinv_1)
                  2.20    5.31    9.37 v _8062_/Y (sky130_fd_sc_hd__clkinv_1)
   260    0.45                           _3834_ (net)
                  2.20    0.00    9.37 v _9277_/B1 (sky130_fd_sc_hd__o21ai_0)
                  0.39    0.38    9.75 ^ _9277_/Y (sky130_fd_sc_hd__o21ai_0)
     1    0.00                           _4573_ (net)
                  0.39    0.00    9.75 ^ _9278_/B1 (sky130_fd_sc_hd__a21oi_1)
                  0.18    0.06    9.81 v _9278_/Y (sky130_fd_sc_hd__a21oi_1)
     1    0.00                           _874_ (net)
                  0.18    0.00    9.81 v _10233_/D (sky130_fd_sc_hd__dfxtp_1)
                                  9.81   data arrival time

                  0.00   11.00   11.00   clock clk (rise edge)
                          0.00   11.00   clock network delay (ideal)
                          0.00   11.00   clock reconvergence pessimism
                                 11.00 ^ _10233_/CLK (sky130_fd_sc_hd__dfxtp_1)
                         -0.16   10.84   library setup time
                                 10.84   data required time
---------------------------------------------------------------------------
                                 10.84   data required time
                                 -9.81   data arrival time
---------------------------------------------------------------------------
                                  1.02   slack (MET)
```

# 6. Observations and Conclusion

STA was completed for all 16 SKY130 HD standard-cell timing corners. The FF and TT corners shown in the captured reports have positive minimum- and maximum-path slack. The SS corners include negative maximum-path slack values, indicating setup-timing violations under those PVT conditions.

The report therefore documents the timing behavior of the same synthesized VSDBabySoC design under the complete 16-corner set used for this assessment.

# 7. Verification Checklist

- 16 SKY130 timing corners included.
- Minimum and maximum timing paths reported for each corner.
- Required report_checks fields used: nets, cap, slew, input_pins and fanout.
- AVSDPLL and AVSDDAC libraries included in the STA flow.
- Synthesized VSDBabySoC netlist and project SDC used consistently.
- Negative slack values are preserved as actual STA results rather than omitted.
- All 16 supplied screenshots are included in the report.

## WNS (Worst Negative Slack) – Setup / Max

WNS Setup / Max values across the 16 SKY130 timing library corners.

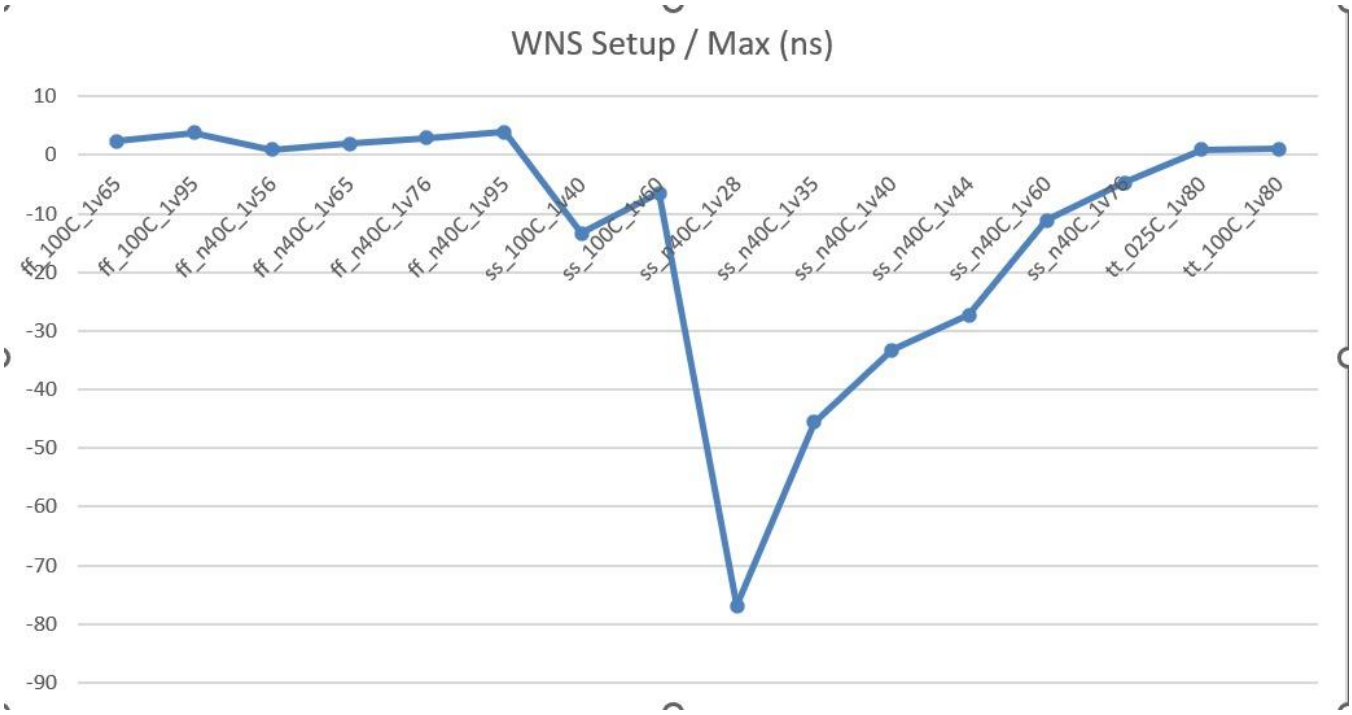



*Figure: WNS Setup / Max across all 16 library corners*

**Note:** The graph was added as an STA results visualization. TNS is not included here because the actual OpenSTA report_tns values have not yet been collected.